CATILLION
Timéo

TD2 MATPLOTLIB

```
from pylab import *
from matplotlib import *

def triangle():
  xlim(0, 10)
  ylim(0, 10)
  x = [2, 5, 8, 2]
  y = [2, 8, 2, 2]
  plot(x, y, '-*r')
  show()

triangle()


def rectangle():
  xlim(0, 10)
  ylim(0, 10)
  x = [3, 3, 7, 7, 3]
  y = [3, 7, 7, 3, 3]
  plot(x, y, '-*r')
  show()

rectangle()


def pointLine():
  xlim(0, 20)
  ylim(2.0, 4.0)
  for i in range(10):
    x = 5 + i
    y = 3
    plot(x, y, '-*b')
  show()

pointLine()


def pointLine2():
  xlim(0, 20)
  ylim(2.0, 6.0)
  for i in range(10):
```

```
    x = 5 + i
    y = 3.0
    plot(x, y, '-*b')
  for i in range(10):
    x = 5 + i
    y = 5.0
    plot(x, y, '-*b')
  show()

pointLine2()


def pointLine3():
  xlim(0, 20)
  ylim(0, 30)
  y = 10
  for i in range(10):
    for j in range(10):
      x = 5 + j
      plot(x, y, '-*b')
    y += 1
  show()

pointLine3()


def affine():
  xlim(0, 50)
  ylim(0, 50)
  x = 10
  y = 10
  for i in range(30):
    x += 1
    y += 1
    plot(x, y, '-*b')
  show()

affine()

def affines():
  xlim(0, 25)
  ylim(0, 45)
  x = 0
  y = 0
  while x < 20:
    x += 1
    y = 2*x
```

```
    plot(x, y, '-*r')
  x = 0
  y = 0
  while x < 20:
    x += 1
    y += 1
    plot(x, y, '-*b')
  x = 0
  y = 0
  while x < 20:
    x += 1
    y = 0.5*x
    plot(x, y, '-*r')
  show()

affines()

def cube():
  xlim(-3, 3)
  ylim(0, 10)
  x = -3
  y = -3
  while x < 3:
    x += 6/100
    y = x**2
    plot(x, y, '-*b')
  show()

cube()

def inverse():
  xlim(-5, 5)
  ylim(-20, 20)
  x = -5
  y = -5
  while x < 5:
    x += 10/500
    y = 1/x
    plot(x, y, '-*b')
  show()

inverse()
```